AWS CLOUD & DATA ENGINEERING ASSIGNMENT

Name : Kushi NC

Roll Number : 23BTRCB018

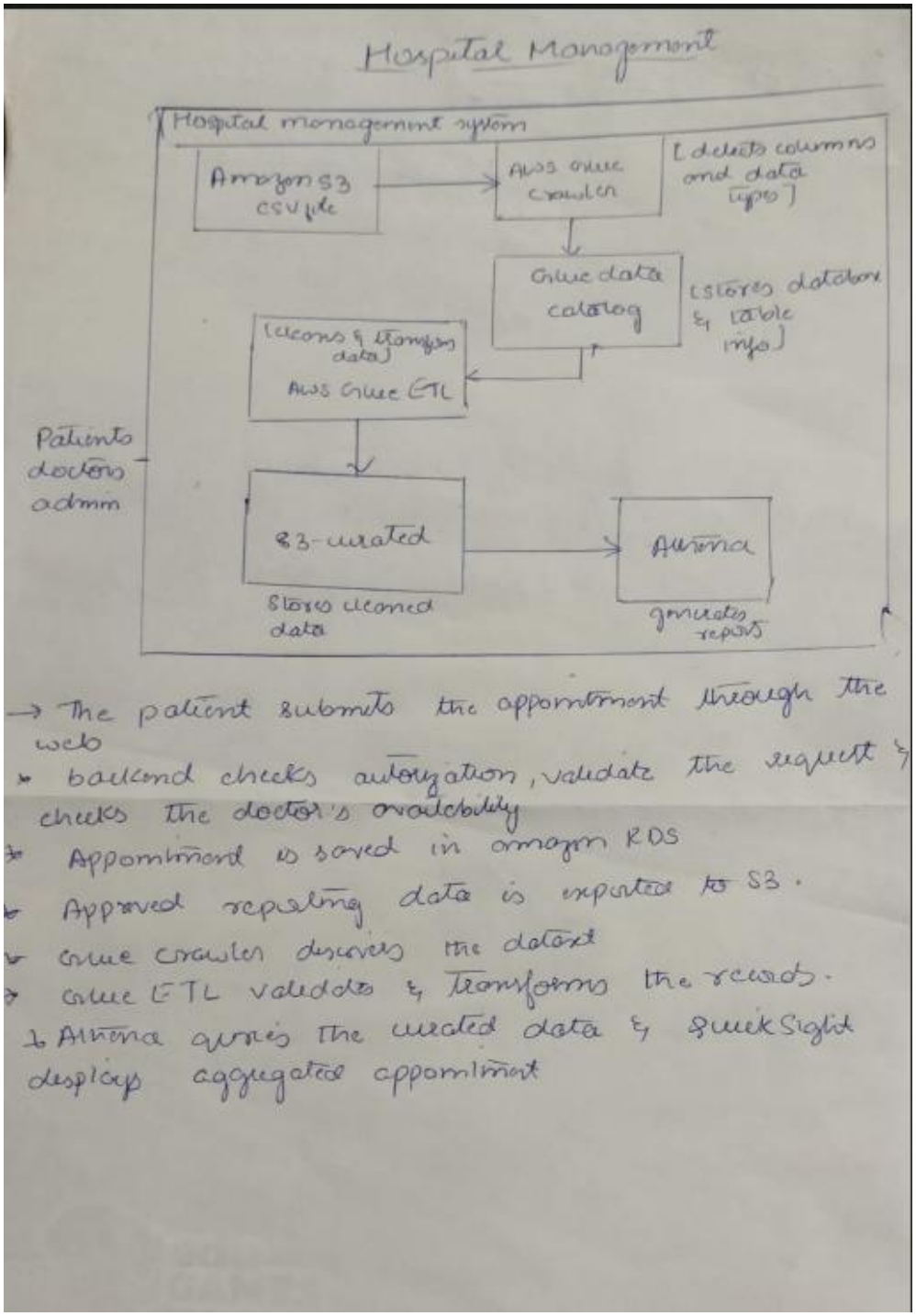
College Name : Jain University

Batch : 2027

Date : 10-10-2026

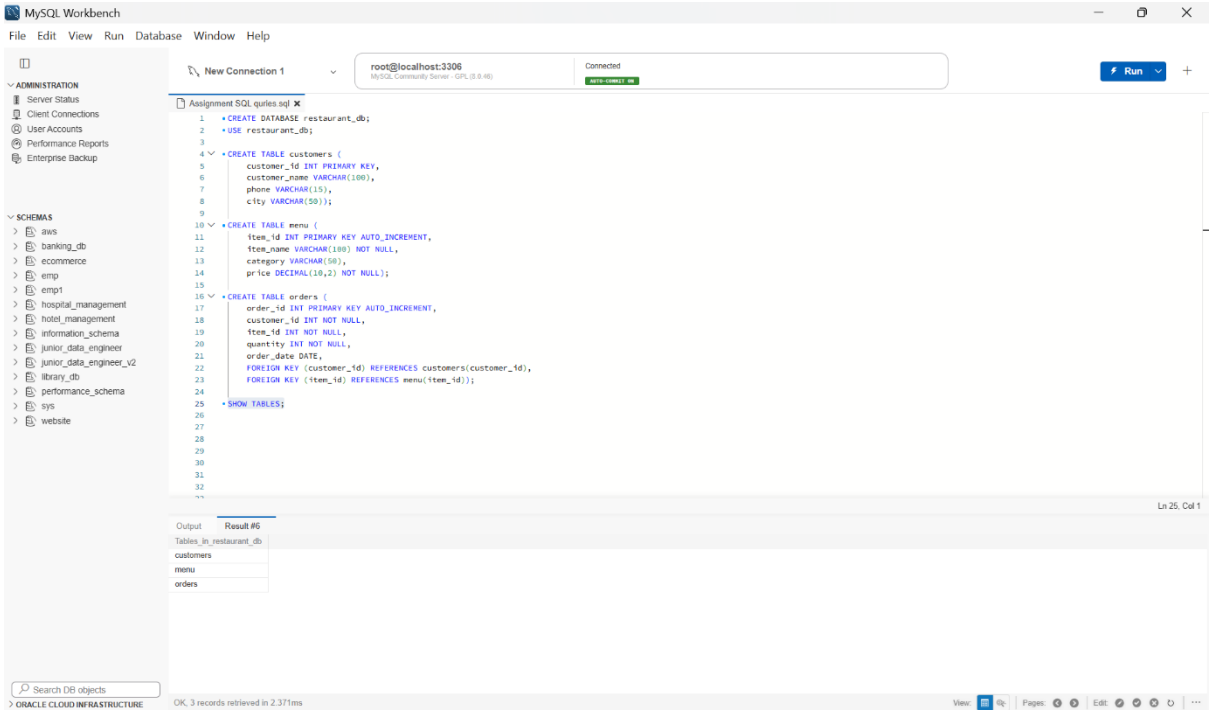
Title : AWS Cloud & Data Engineering

Hotel Management Architecture

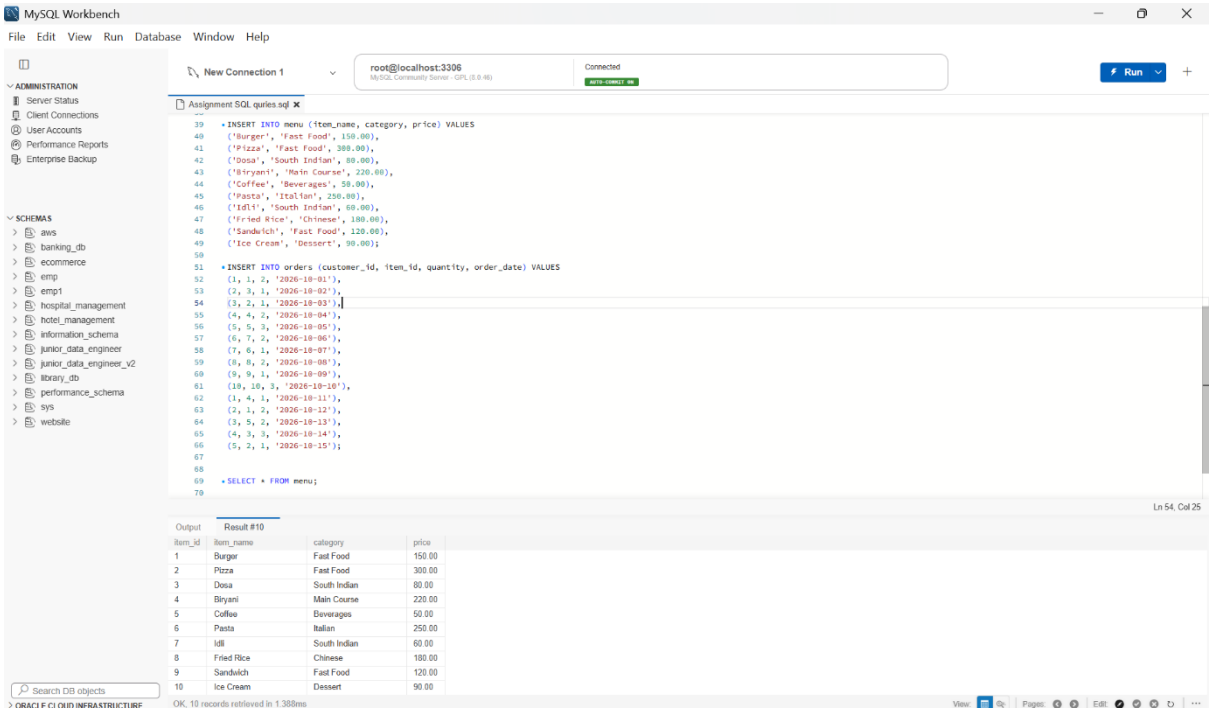


→ The patient submits the appointment through the web
* backend checks authorization, validate the request & checks the doctor's availability
* Appointment is saved in amazon RDS
* Approved repeating data is exported to S3.
* Glue Crawler discovers the dataset
* Glue ETL validates & transforms the records.
* Athena query's the curated data & QuickSight display aggregated appointment

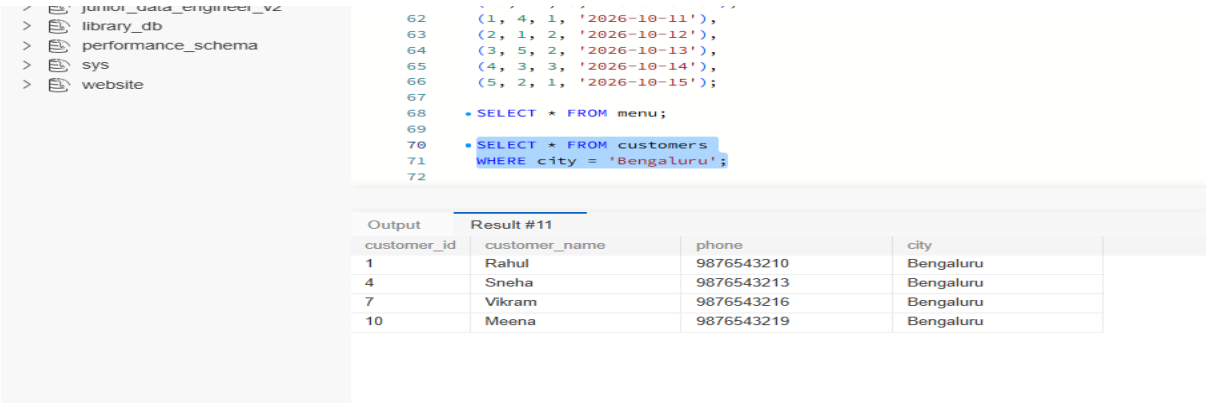
1. Create a Table in MySQL



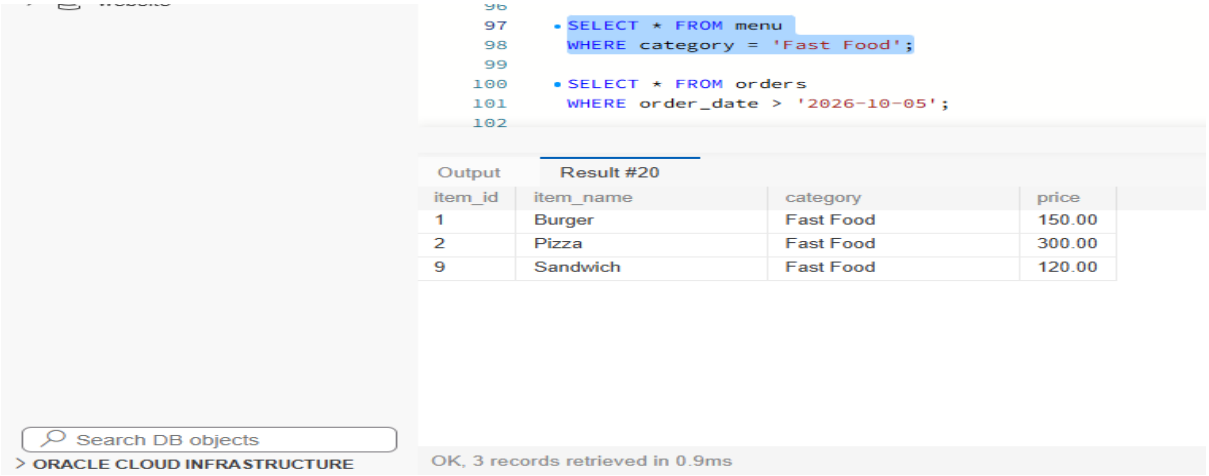
2. Inserting Values



1. Customer details

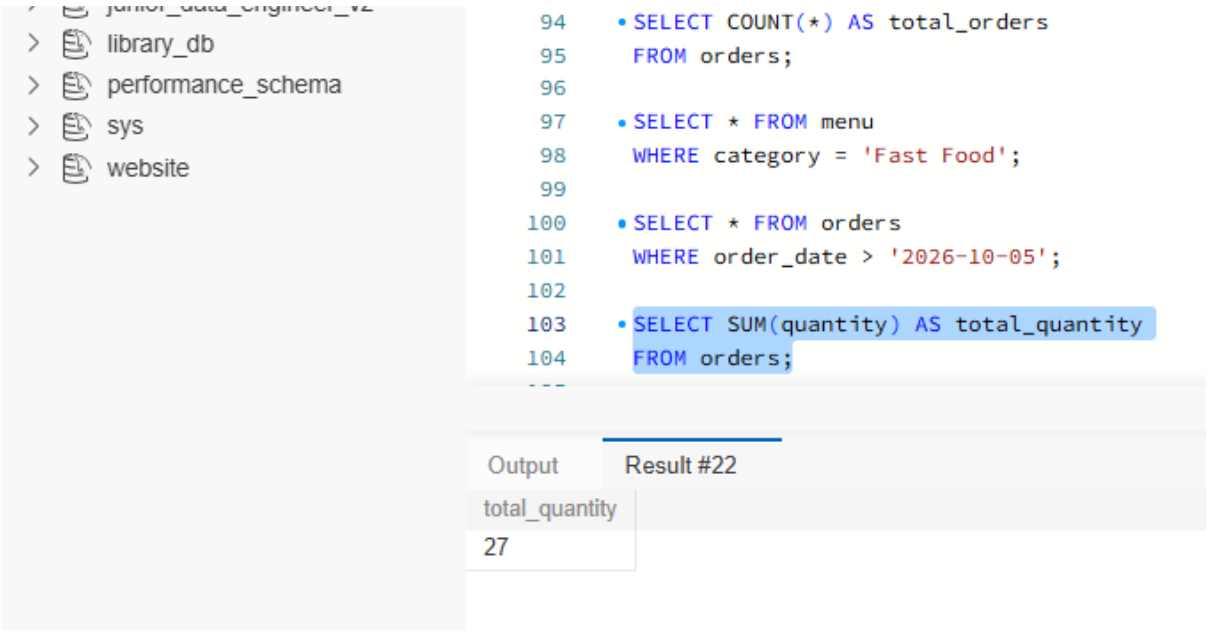


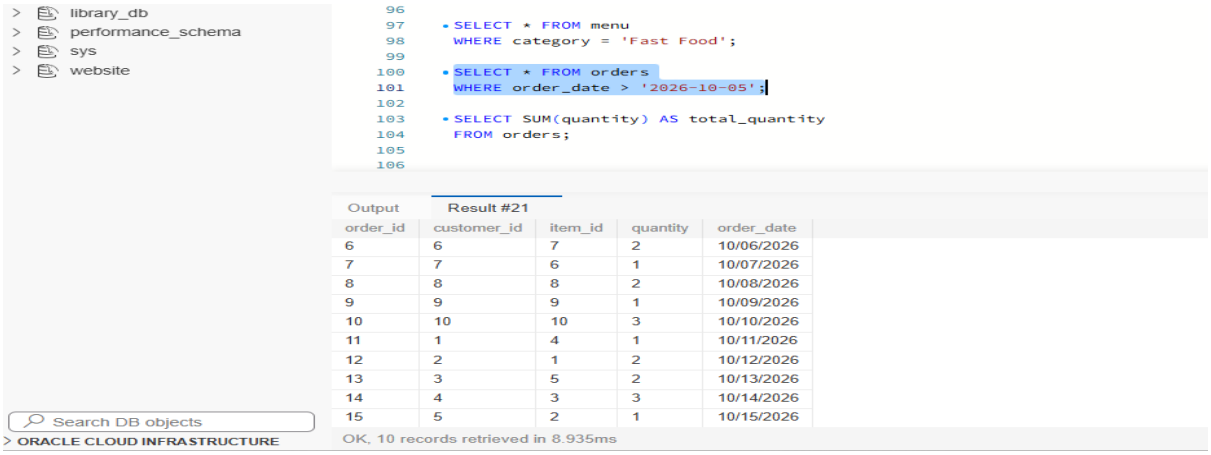
2. Menu items



3.Total quantity

4. Order details

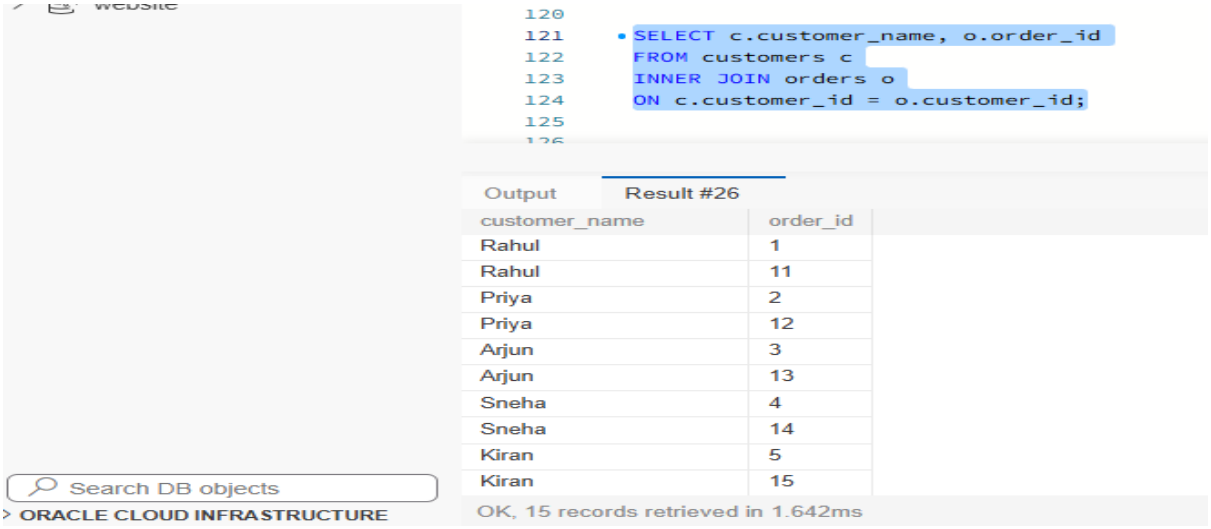




5. Item quantity



6. Customer orders



7. Customer order details

website

```
110
111  SELECT c.customer_name, o.order_id
112  FROM customers c
113  LEFT JOIN orders o
114  ON c.customer_id = o.customer_id;
115
```

Output | Result #24

customer_name	order_id
Rahul	1
Rahul	11
Priya	2
Priya	12
Arjun	3
Arjun	13
Sneha	4
Sneha	14
Kiran	5
Kiran	15

Search DB objects

ORACLE CLOUD INFRASTRUCTURE

OK, 15 records retrieved in 2.451ms

8. Item names and quantities.

website

```
105
106  SELECT m.item_name, o.quantity
107  FROM menu m
108  INNER JOIN orders o
109  ON m.item_id = o.item_id;
110
111
```

Output | Result #23

item_name	quantity
Burger	2
Burger	2
Pizza	1
Pizza	1
Dosa	1
Dosa	3
Biryani	2
Biryani	1
Coffee	3
Coffee	2

Search DB objects

ORACLE CLOUD INFRASTRUCTURE

OK, 15 records retrieved in 2.02ms

9. Customer names

website

```
76   SELECT * FROM customers
77   ORDER BY customer_name ASC;
125
126  SELECT c.city, COUNT(o.order_id) AS total_orders
127  FROM customers c
128  LEFT JOIN orders o
129  ON c.customer_id = o.customer_id
130  GROUP BY c.city;
131
```

Output | Result #27

city	total_orders
Bengaluru	6
Mysuru	3
Mangaluru	3
Hubballi	2
Udupi	1

7	Vikram	9876543216	Bengaluru

Search DB objects

ORACLE CLOUD INFRASTRUCTURE

OK, 10 records retrieved in 1.863ms

10. Orders by city

11. Customers sorted by name.

```
76  SELECT * FROM customers
77  ORDER BY customer_name ASC;
78
```

Output | Result #13

customer_id	customer_name	phone	city
6	Ananya	9876543215	Mysuru
3	Arjun	9876543212	Mangaluru
8	Divya	9876543217	Udupi
5	Kiran	9876543214	Hubballi
10	Meena	9876543219	Bengaluru
2	Priya	9876543211	Mysuru
1	Rahul	9876543210	Bengaluru
9	Rohit	9876543218	Mangaluru
4	Sneha	9876543213	Bengaluru
7	Vikram	9876543216	Bengaluru

Search DB objects

> ORACLE CLOUD INFRASTRUCTURE

OK, 10 records retrieved in 1.863ms

12. Menu prices above average

```
124  GROUP BY c.city;
125
126  SELECT item_name, price
127  FROM menu
128  WHERE price > (SELECT AVG(price) FROM menu);
129
```

Output | Result #26

item_name	price
Pizza	300.00
Biryani	220.00
Pasta	250.00
Fried Rice	180.00

13. Highest-priced item

```
129
130  SELECT item_name, category, price
131  FROM menu
132  WHERE price = (SELECT MAX(price) FROM menu);
133
```

Output | Result #27

item_name	category	price
Pizza	Fast Food	300.00

14. Customer order summary

website

```
133
134  SELECT c.customer_name, m.item_name, o.quantity, o.order_date
135  FROM customers c
136  JOIN orders o ON c.customer_id = o.customer_id
137  JOIN menu m ON o.item_id = m.item_id;
138
139
```

Output | Result #28

customer_name	item_name	quantity	order_date
Rahul	Burger	2	10/01/2026
Rahul	Biryani	1	10/11/2026
Priya	Dosa	1	10/02/2026
Priya	Burger	2	10/12/2026
Arjun	Pizza	1	10/03/2026
Arjun	Coffee	2	10/13/2026
Sneha	Biryani	2	10/04/2026
Sneha	Dosa	3	10/14/2026
Kiran	Coffee	3	10/05/2026
Kiran	Pizza	1	10/15/2026

Search DB objects

ORACLE CLOUD INFRASTRUCTURE

OK, 15 records retrieved in 1.926ms

15. Total items by category

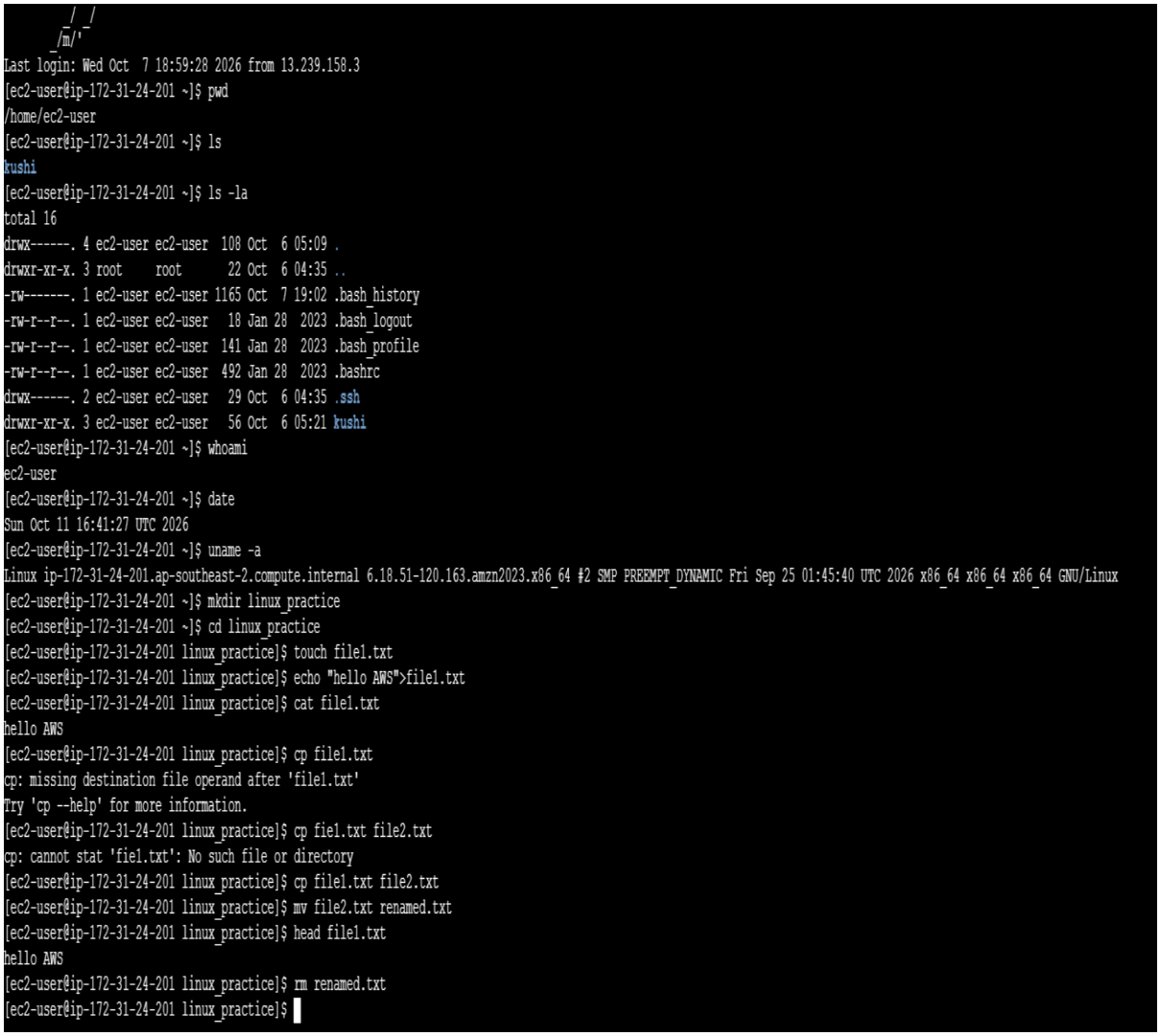
website

```
138
139  SELECT m.category, SUM(o.quantity) AS total_quantity
140  FROM menu m
141  INNER JOIN orders o
142  ON m.item_id = o.item_id
143  GROUP BY m.category;
144
```

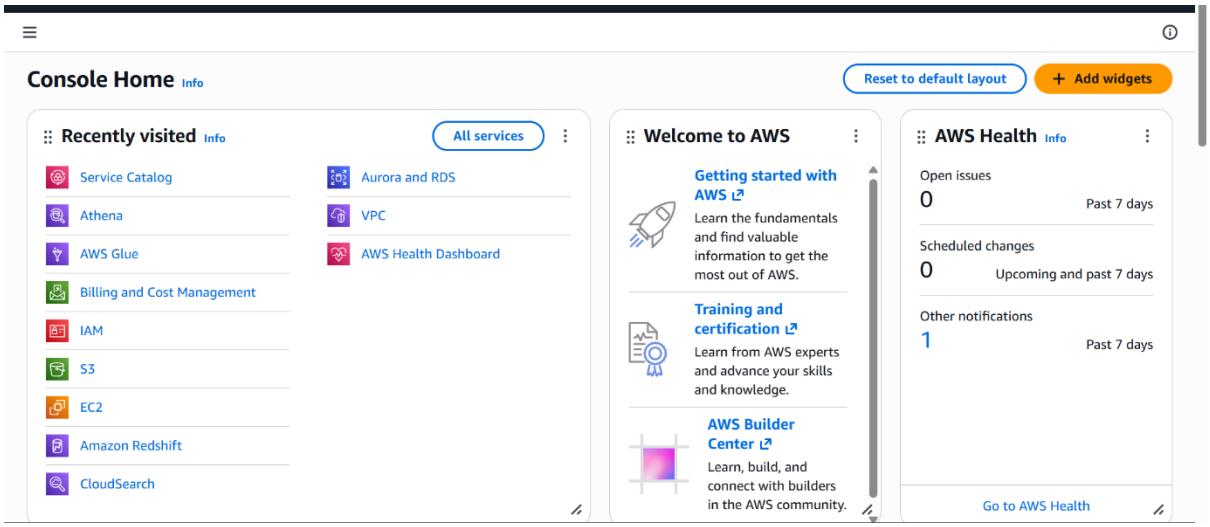
Output | Result #29

category	total_quantity
Fast Food	7
South Indian	6
Main Course	3
Beverages	5
Italian	1
Chinese	2
Dessert	3

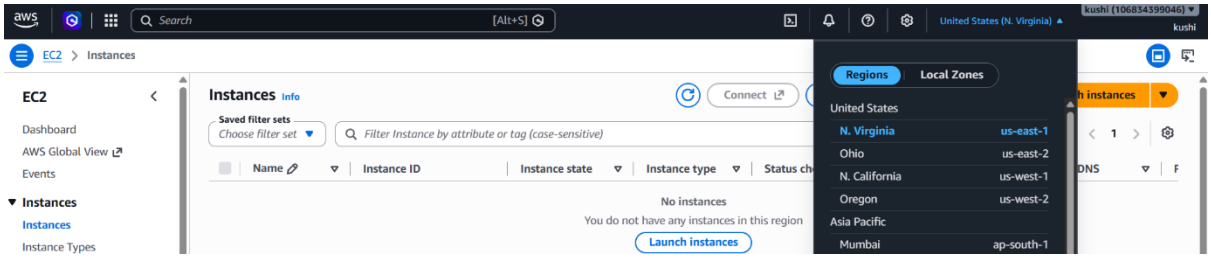
Linux Commands

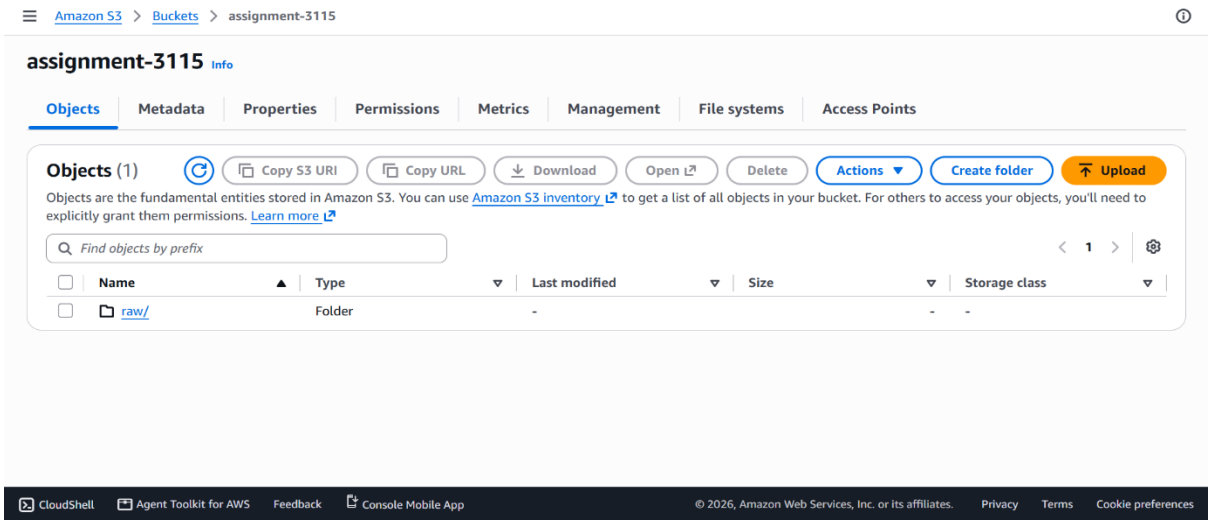


AWS Account / Region

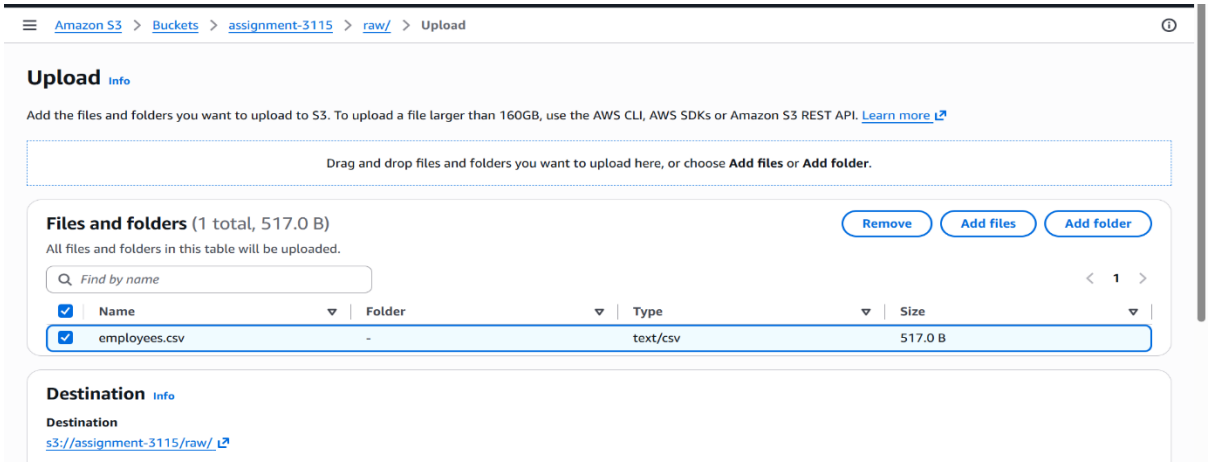


AWS Region : us-east-1

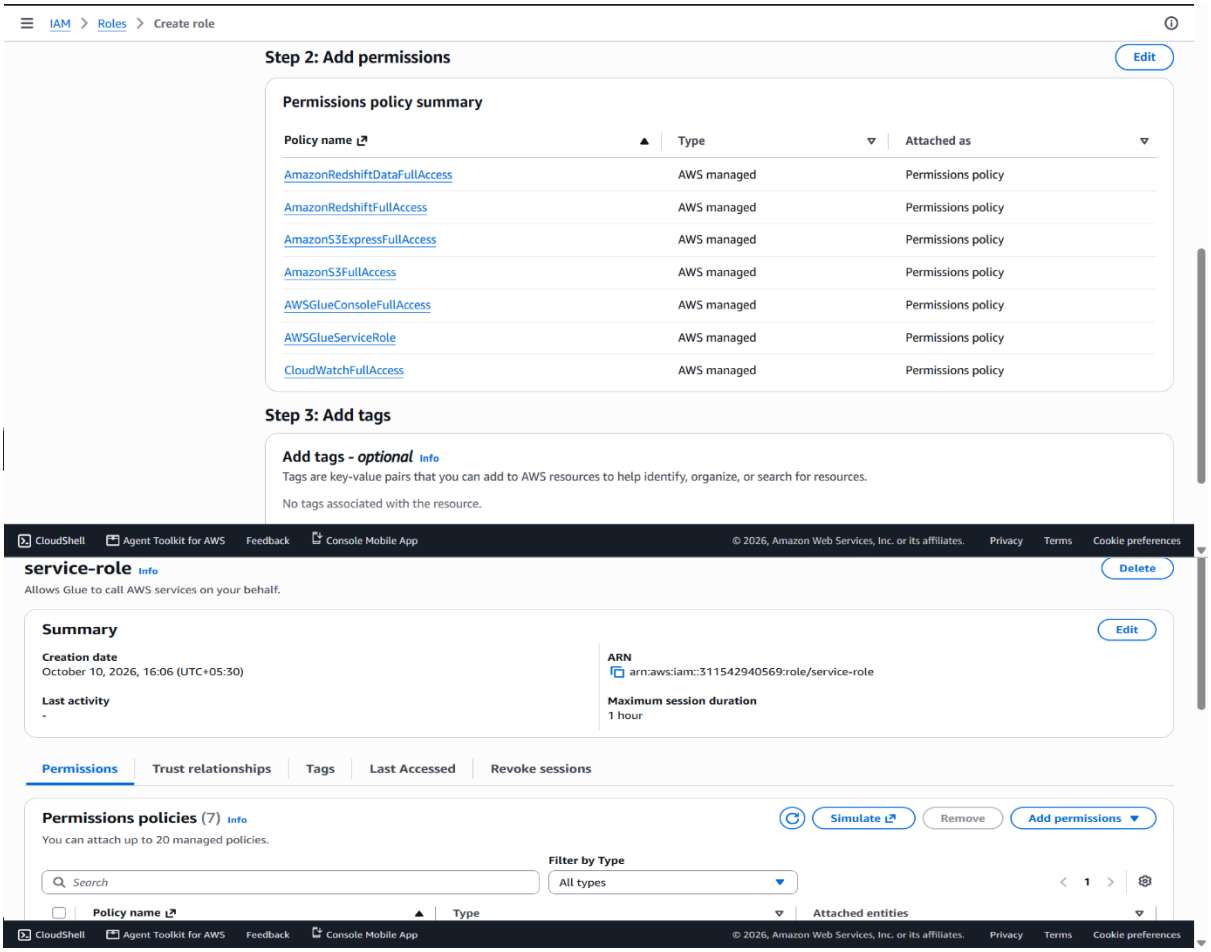




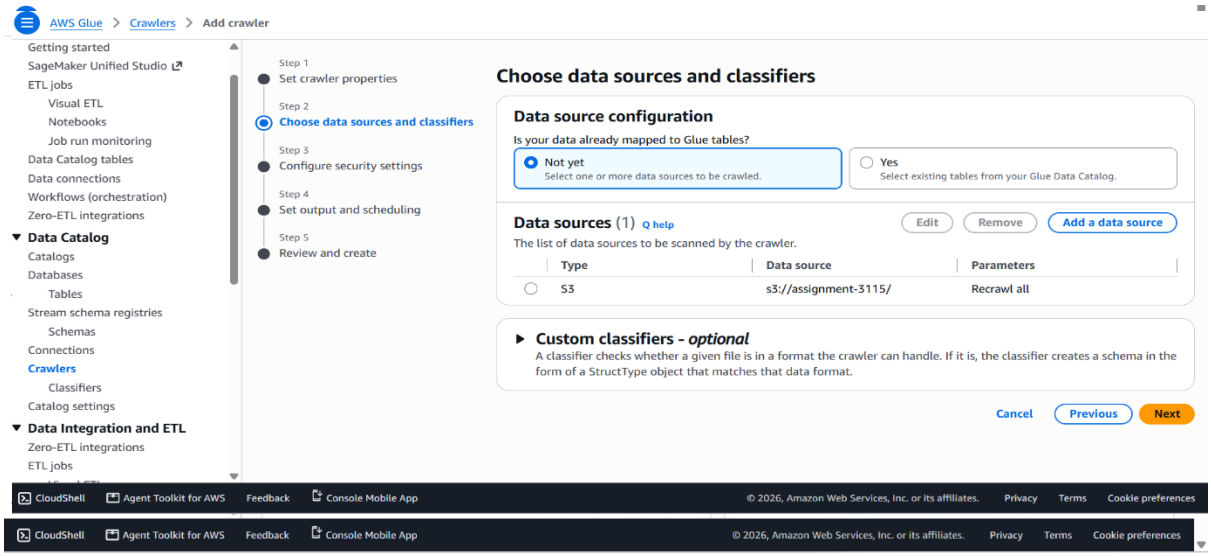
Amazon S3



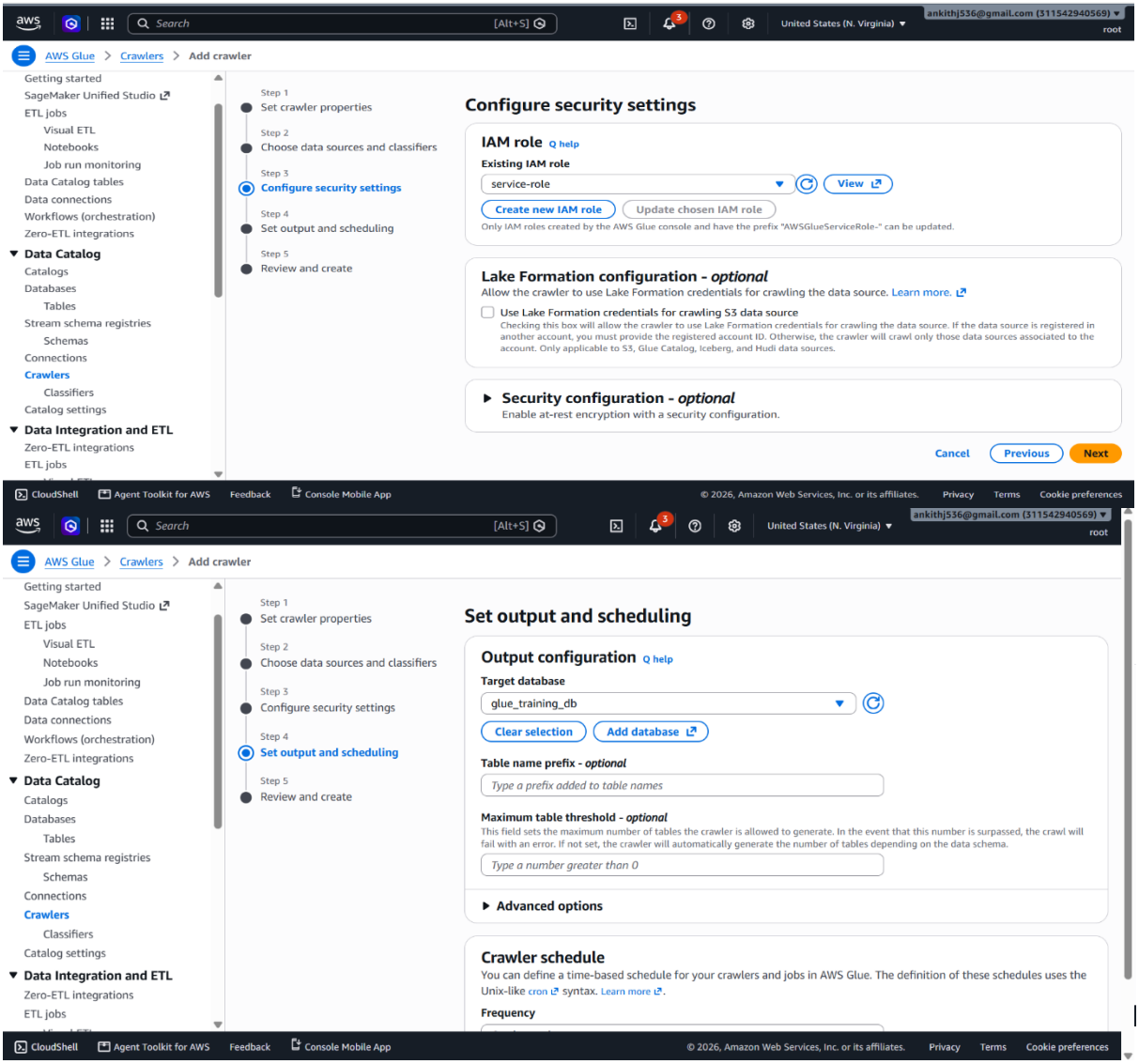
IAM (Identity and Access Management)

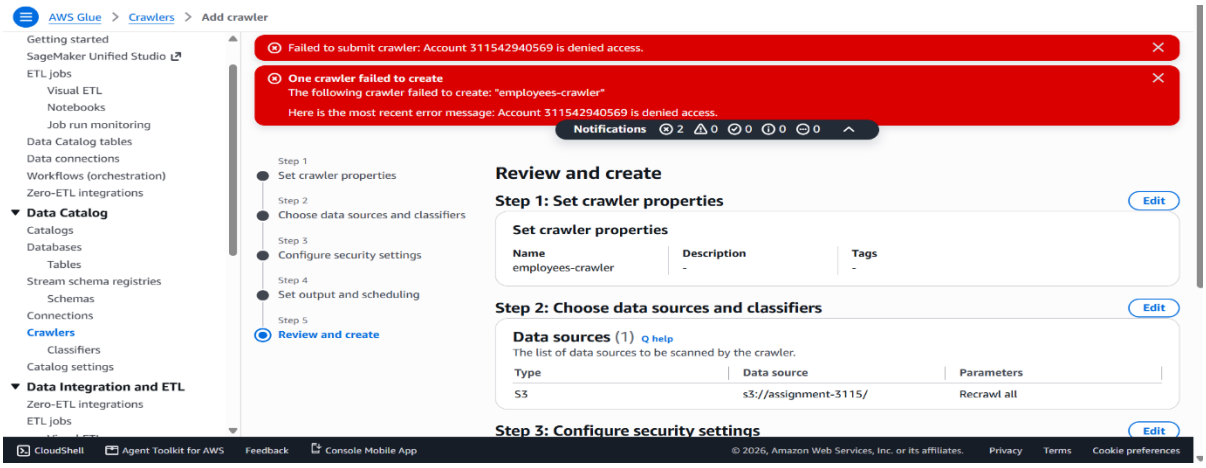


AWS Glue Database

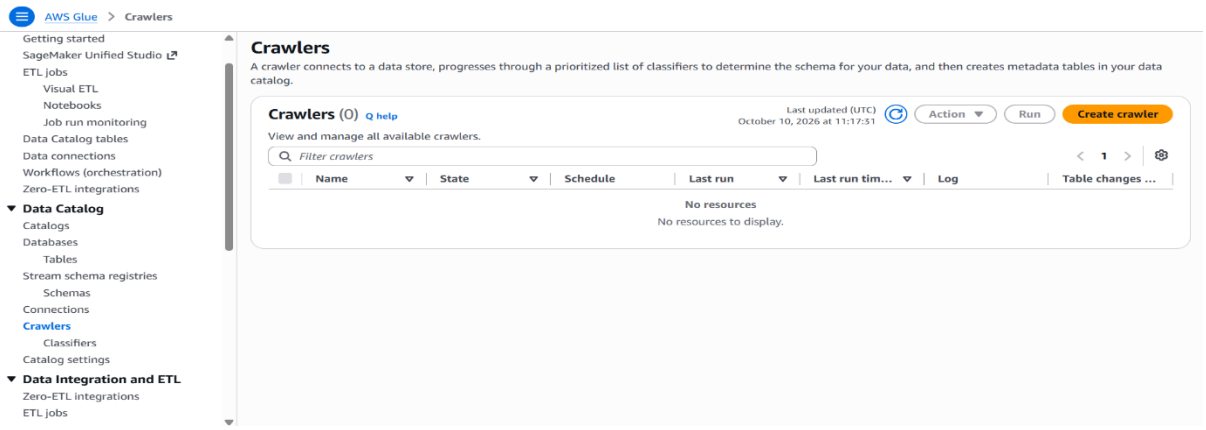


Glue Crawler

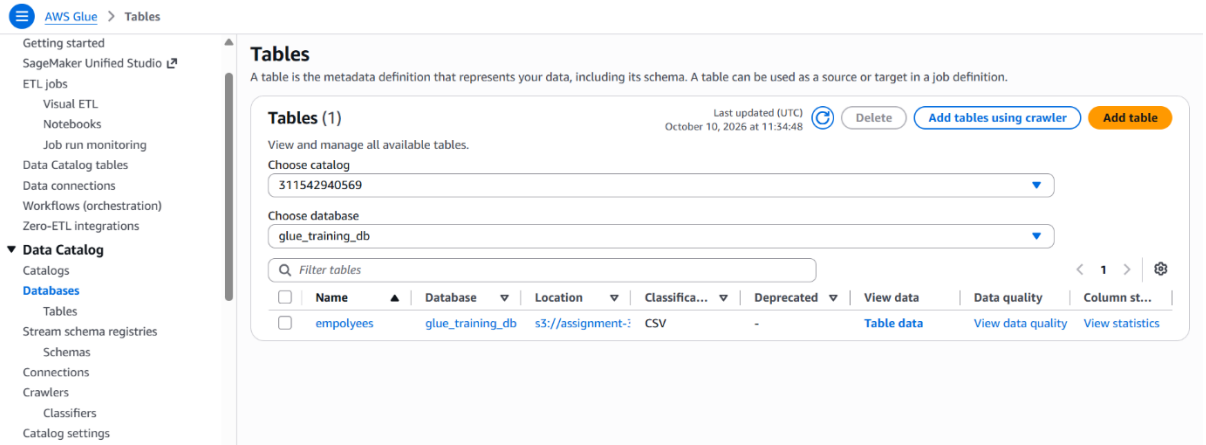


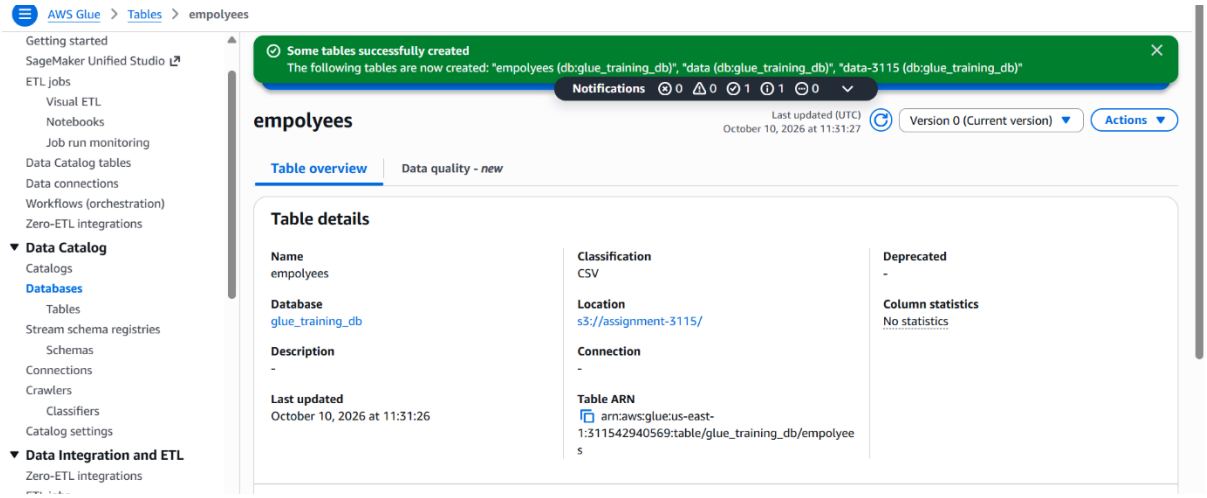


Crawler Execution

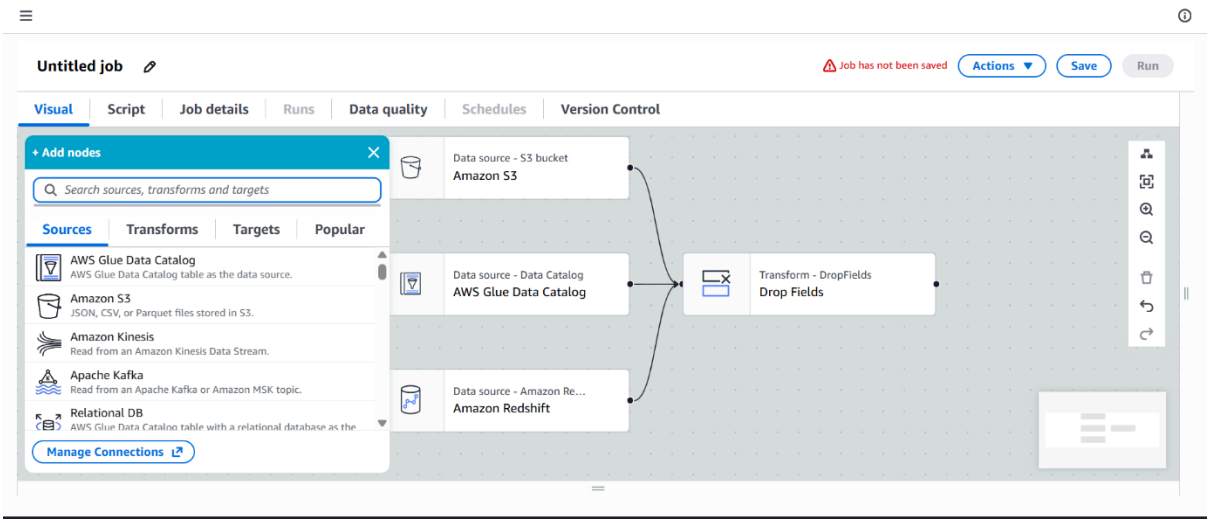


Glue Data Catalog

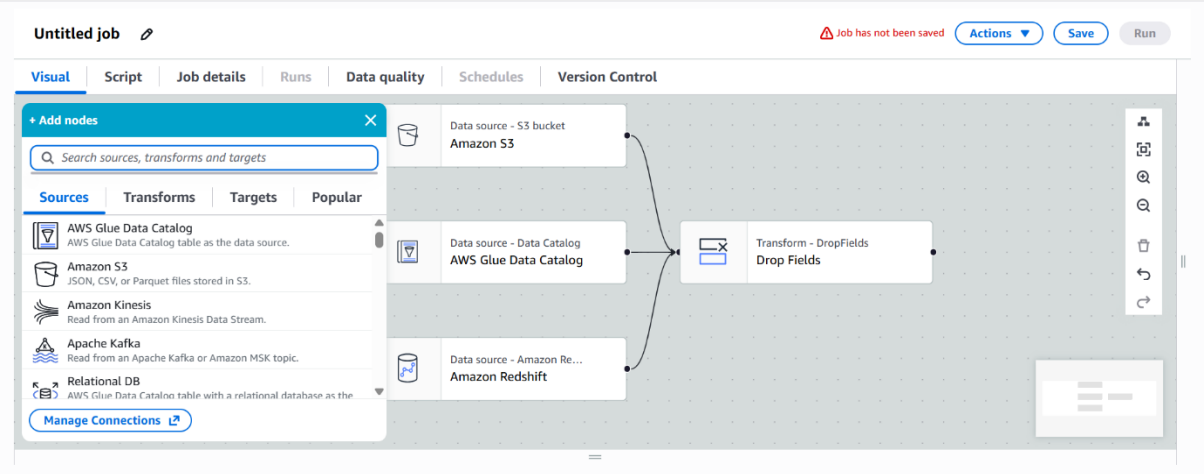




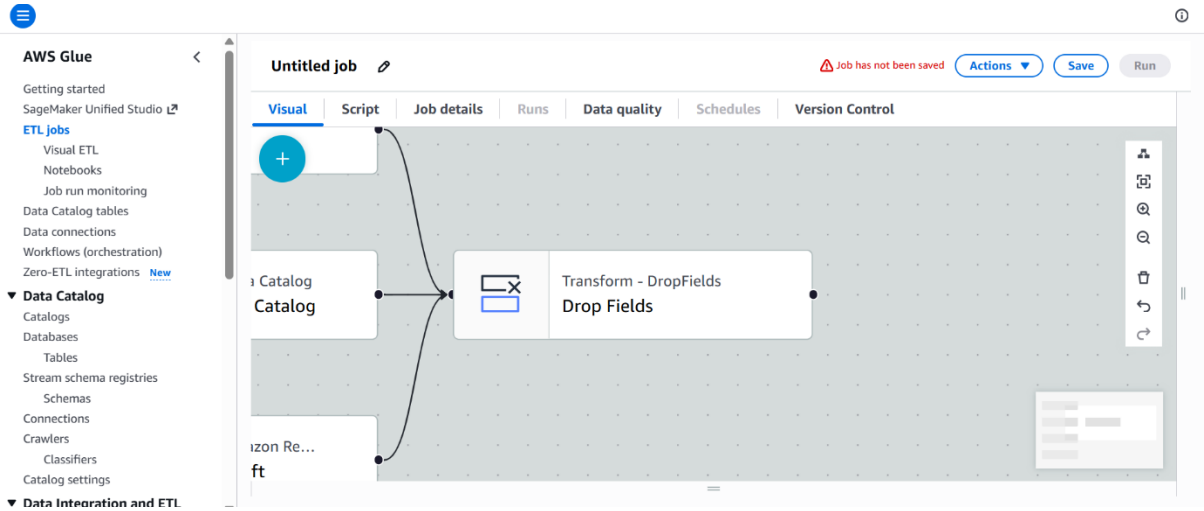
Glue Visual ETL



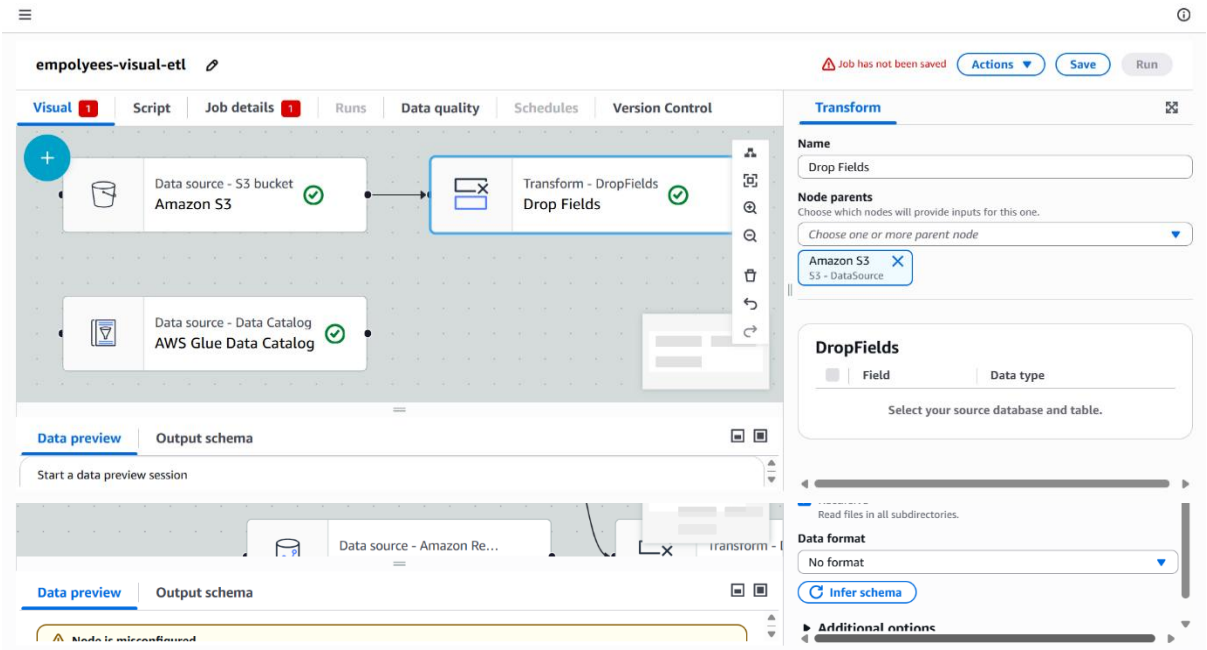
Glue Transformation



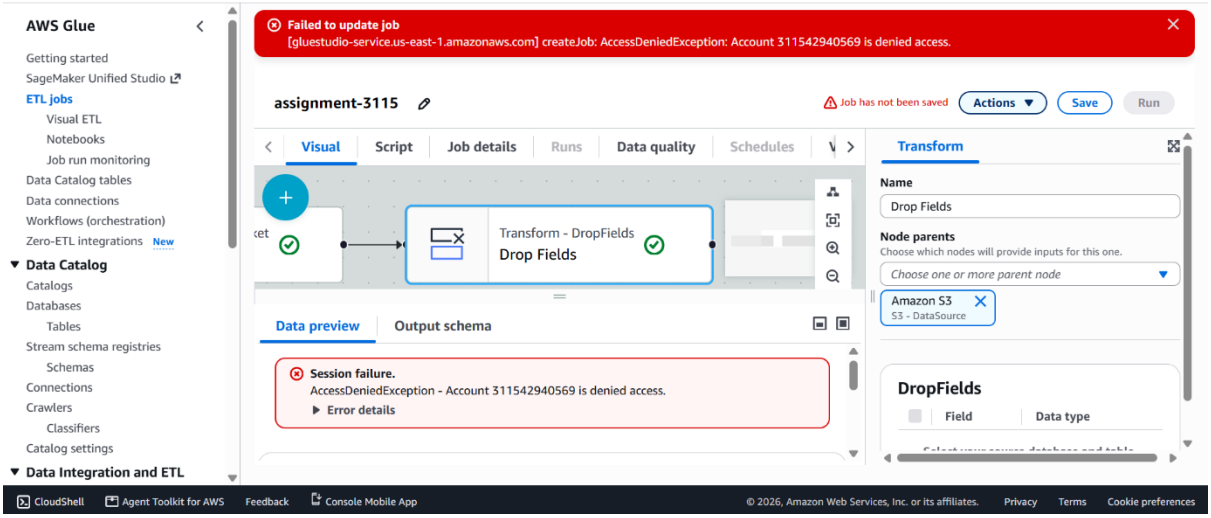
Drop fields



S3 Target Configuration



Job status = Succeeded/Denied



Explainations:

1. Amazon S3

Amazon S3 is used to store the data. We created a bucket with two folders called raw and curated. The original file is uploaded in the raw folder. The curated folder is used to store the cleaned data.

2. Glue Crawler

The Glue crawler scans the data present in the S3 raw folder. It identifies the file format, column names and data types.

3. Glue Data Catalog

Glue Data Catalog stores the information about the database and table.

4. Glue Visual ETL

We connect the source table, transformation and S3 target. It helps us process the data without writing the complete code manually.

5. S3 Curated Data

The data is stored in the curated folder in S3 .When the ETL job runs successfully, the output files are created in the curated folder. This cleaned data can be used for further analysis and reporting.